Experiment	Type	Qwentaur 0.6B	Qwentaur 1.7B	Qwentaur 4B	Qwentaur 8B	Qwentaur 14B	Llama-Centaur 1B	Llama-Centaur 3B	Llama-Centaur 8B	Smoltaur 0.1B	Smoltaur 0.4B	Smoltaur 1.7B	Smoltaur 3B	Olmotaur 1B	Olmotaur 7B	[binz2025foundation] L-70B$_p$	[binz2025foundation] C$_r$	[binz2025foundation] C$_p$	[binz2025foundation] Cog.$_p$	Chance ln(k)
Balloon analog risk task [frey2017risk]	Decision	0.07	0.07	0.07	0.06	0.06	0.07	0.07	**0.06**	0.08	0.07	0.07	0.07	0.07	0.07	0.08	0.07	0.06	0.09	0.69
CPC18 [plonsky2018when]	Decision	0.35	0.34	0.34	0.34	**0.33**	0.36	0.35	0.34	0.42	0.41	0.40	0.34	0.44	0.34	0.41	0.35	0.34	0.66	0.69
Columbia card task [frey2017risk]	Decision	0.20	0.20	0.20	0.20	0.20	0.21	0.20	**0.20**	0.24	0.22	0.21	0.20	0.22	0.20	0.23	0.21	0.19	0.26	0.69
Decisions from description [wulff2018sampling]	Decision	0.62	0.63	0.62	0.61	0.60	0.62	0.63	**0.60**	0.68	0.66	0.64	0.61	0.63	0.62	0.76	0.59	0.53	0.61	0.69
Experiential-symbolic task [garcia2023experiential]	Decision	0.48	0.47	0.46	0.46	0.46	0.47	0.47	0.46	0.47	0.45	**0.43**	0.47	0.47	0.47	0.70	0.46	0.45	—	—
Gardening task [flesch2018comparing]	Decision	0.39	0.39	0.38	0.38	0.38	0.40	0.39	**0.38**	0.57	0.54	0.53	0.38	0.58	0.38	0.50	0.49	0.38	0.91	0.69
Go/no-go [enkavi2019large]	Decision	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.08	0.00
Intertemporal choice [ruggeri2022globalizability]	Decision	0.44	0.44	**0.44**	0.44	0.44	0.44	**0.44**	0.44	0.47	0.45	0.44	0.44	0.44	0.44	0.73	0.44	0.43	0.66	0.69
Multi-attribute DM [hilbig2014generalized]	Decision	0.09	0.08	0.09	0.08	0.08	0.08	0.08	0.08	0.31	0.19	0.16	**0.08**	0.10	0.09	0.15	0.06	0.06	0.19	0.69
Risky choice [krueger2024identifying]	Decision	0.48	0.47	0.44	0.43	**0.42**	0.49	0.46	0.43	0.72	0.61	0.53	0.45	0.58	0.46	0.65	0.43	0.43	—	—
choices13k [peterson2021using]	Decision	0.44	0.44	0.43	0.43	**0.43**	0.44	0.44	0.43	0.49	0.46	0.44	0.44	0.45	0.44	0.53	0.43	0.43	0.66	0.69
Multi-task RL [tomov2021multitask]	MDP	0.60	0.59	0.58	0.58	**0.57**	0.60	0.59	0.57	0.68	0.66	0.63	0.59	0.78	0.59	0.66	0.57	0.57	1.04	1.10
Tile-revealing task [kumar2023disentangling]	MDP	0.02	0.01	0.00	0.07	0.00	0.02	0.00	**0.00**	2.69	1.99	0.04	0.00	0.06	0.09	2.74	2.29	1.87	—	—
Two-step task [kool2017cost]	MDP	0.54	0.53	**0.53**	0.53	**0.53**	0.55	0.53	**0.53**	0.60	0.60	0.56	0.53	0.57	0.53	0.61	0.53	0.50†	0.60†	0.69
Two-step task [kool2016when]	MDP	0.51	0.50	0.49	0.49	**0.48**	0.52	0.50	0.49	0.59	0.58	0.52	0.50	0.55	0.50	0.61	0.48	0.50†	0.60†	0.69
Two-step task [zorowitz2023data]	MDP	0.52	0.51	0.51	**0.51**	0.51	0.52	0.51	0.51	0.60	0.58	0.52	0.52	0.54	0.51	0.61	0.51	0.50†	0.60†	0.69
Virtual subway network [tomov2020discovery]	MDP	1.16	1.15	1.14	1.13	**1.12**	1.19	1.16	1.13	1.43	1.33	1.30	1.14	1.32	1.15	1.53	1.16	1.13	—	1.61
Zoopermarket [ludwig2023human]	MDP	0.54	0.52	0.51	0.50	**0.48**	0.54	0.52	0.50	0.63	0.61	0.56	0.51	0.61	0.52	0.60	0.50	0.48	0.60	0.69
Cond. assoc. learning [collins2014working]	Memory	0.54	0.53	0.52	0.52	**0.51**	0.55	0.53	0.52	0.74	0.64	0.58	0.53	0.65	0.53	0.64	0.52	0.54	0.86	1.10
Digit span [enkavi2019large]	Memory	0.60	0.59	0.58	0.57	**0.57**	0.60	0.59	0.58	0.71	0.65	0.62	0.59	0.62	0.59	0.66	0.57	0.55	0.94	2.30
Episodic long-term memory [popov2023intent]	Memory	1.27	1.05	0.96	0.94	**0.90**	1.28	1.02	0.92	1.88	1.56	1.37	1.01	1.23	1.05	1.13	0.89	0.87	—	0.69
N-back [enkavi2019large]	Memory	0.42	0.42	0.41	0.41	**0.40**	0.42	0.42	0.41	0.49	0.47	0.44	0.41	0.44	0.42	0.52	0.40	0.40	0.58	0.69
Recall and recognition [cox2018information]	Memory	1.24	1.18	1.13	1.12	**1.09**	1.22	1.14	1.10	1.69	1.50	1.25	1.15	1.41	1.17	1.38	1.07	1.06	—	—
Recent probes [enkavi2019large]	Memory	0.26	0.26	0.26	0.26	0.26	0.27	0.26	**0.26**	0.33	0.28	0.27	0.26	0.27	0.26	0.34	0.26	0.26	0.39	0.69
Grammar judgement [jansen2021rational]	Misc.	1.47	1.45	1.45	1.44	**1.43**	1.48	1.45	1.44	1.63	1.54	1.51	1.46	1.51	1.46	1.99	1.44	1.44	1.41	—
Probabilistic reasoning [zhu2020bayesian]	Misc.	2.45	2.40	2.36	2.35	**2.34**	2.56	2.47	2.43	2.67	2.60	2.50	2.45	2.61	2.47	2.64	2.40	2.37	—	—
Serial reaction time task [wu2023chunking]	Misc.	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.16	**0.16**	0.17	0.16	0.17	0.19	0.18	0.17	0.20	1.39
THINGS odd-one-out [hebart2023things]	Misc.	0.82	0.81	0.80	0.79	**0.79**	0.82	0.82	0.79	1.08	0.90	0.86	0.80	0.86	0.81	1.14	0.80	0.81	0.83	1.10
Changing bandit [xiong2023neural]	Bandit	0.30	0.30	0.29	0.29	**0.29**	0.32	0.31	0.29	0.40	0.38	0.35	0.30	0.39	0.30	0.38	0.45	0.30	0.44	0.69
Decisions from experience [wulff2018sampling]	Bandit	0.49	0.49	0.49	0.49	0.48	0.50	0.49	**0.48**	0.54	0.52	0.50	0.49	0.50	0.49	0.43	0.47	0.37	0.54	—
Drifting four-armed bandit [bahrami2020four]	Bandit	0.73	0.72	0.71	0.71	**0.71**	0.73	0.72	0.71	0.83	0.78	0.74	0.72	0.74	0.72	0.88	0.71	0.70	0.90	1.39
Horizon task [feng2021dynamics]	Bandit	0.23	0.23	0.22	0.22	0.22	0.26	0.24	**0.22**	0.40	0.34	0.30	0.22	0.35	0.22	0.52	0.40	0.40†	0.36†	0.69
Horizon task [sadeghiyeh2020temporal]	Bandit	0.59	0.58	0.58	0.58	**0.58**	0.59	0.58	0.58	0.62	0.62	0.60	0.58	0.61	0.58	0.52	0.58	0.40†	0.36†	0.69
Horizon task [somerville2017charting]	Bandit	0.34	0.34	0.34	0.34	**0.33**	0.35	0.34	0.34	0.48	0.40	0.38	0.34	0.40	0.34	0.52	0.35	0.40†	0.36†	0.69
Horizon task [waltz2020differential]	Bandit	0.15	0.15	0.15	0.15	0.15	0.16	0.16	**0.14**	0.26	0.21	0.18	0.15	0.19	0.15	0.52	0.15	0.40†	0.36†	0.69
Horizon task [wilson2014humans]	Bandit	0.46	0.46	0.45	0.45	**0.45**	0.47	0.46	0.45	0.54	0.51	0.49	0.45	0.52	0.45	0.52	0.48	0.40†	0.36†	0.69
Iowa gambling task [steingroever2015data]	Bandit	0.93	0.93	0.91	0.91	0.91	0.94	0.92	**0.91**	1.01	0.98	0.94	0.92	0.95	0.92	0.99	0.91	0.89	1.16	1.39
Prob. instrumental learning [lefebvre2017behavioural]	Bandit	0.50	0.50	**0.49**	0.49	0.49	0.51	0.50	0.49	0.54	0.53	0.51	0.50	0.51	0.50	0.54	0.50	0.49	0.50	0.69
Spatially correlated MAB [wu2018generalization]	Bandit	2.01	1.96	1.93	1.87	**1.86**	2.16	2.00	1.93	2.50	2.30	2.16	1.95	2.26	2.06	2.45	1.82	1.83	2.76	3.40
Structured bandit [schulz2020finding]	Bandit	0.67	0.66	0.65	0.65	**0.64**	0.67	0.66	0.65	0.77	0.71	0.68	0.65	0.73	0.66	0.81	0.64	0.64	1.05	2.08
Two-armed bandit [gershman2018deconstructing]	Bandit	0.30	0.30	0.29	0.29	**0.29**	0.30	0.29	0.29	0.37	0.33	0.31	0.29	0.31	0.30	0.38	0.30	0.30	0.42	0.69
Aversive learning [wise2019computational]	Sup. learn.	4.71	4.55	4.27	4.24	3.81	4.58	4.80	**3.66**	5.35	5.12	4.92	3.93	5.44	4.67	5.11	4.13	4.07	—	—
Medin categorization [levering2020revisiting]	Sup. learn.	0.51	0.51	0.50	0.50	**0.50**	0.52	0.50	0.50	0.56	0.54	0.52	0.50	0.52	0.51	0.58	0.50	0.50	0.53	—
Multiple-cue judgment [collsioo2023numerical]	Sup. learn.	1.17	1.16	1.14	1.14	1.13	1.17	1.15	**1.12**	1.59	1.49	1.40	1.14	1.67	1.15	1.28	1.14	1.12	1.92	2.20
Shepard categorization [badham2017deficits]	Sup. learn.	0.55	0.55	0.54	0.53	**0.53**	0.56	0.54	0.53	0.62	0.58	0.56	0.54	0.56	0.54	0.58	0.54	0.54	0.61	0.69
Weather prediction task [speekenbrink2008learning]	Sup. learn.	0.55	0.56	0.54	0.55	0.55	0.55	0.54	0.56	0.58	0.56	0.56	0.55	0.57	**0.53**	0.57	0.56	0.55	0.63	0.69
Mean		0.69	0.68	0.66	0.66	**0.64**	0.70	0.68	0.64	0.89	0.82	0.73	0.66	0.77	0.68	0.88	0.71	0.69	0.69	0.98

<0.20 | 0.20–0.40 | 0.40–0.60 | 0.60–0.80 | 0.80–1.00 | 1.00–1.20 | 1.20–1.50 | 1.50–2.00 | >2.00

Bold+underline = best; underline = second-best (across the four families; reference columns excluded). Lower is better. "—" = not available. All models finetuned at LoRA rank 16, bf16 inference. Reference columns ([binz2025foundation]): L-70B$_p$ = Llama-3.1-70B base, C$_r$ = our reproduced Centaur-70B, C$_p$ = published Centaur-70B, Cog.$_p$ = domain-specific cognitive baseline. †Horizon and Two-step tasks are reported as single merged values by [binz2025foundation]. Rounded to 2 d.p.

Table 1: Per-experiment NLL on Psych-101 for all four finetuned families (bf16, LoRA rank 16). Cell colour encodes performance (green = better).

Experiment	Type	Qwen3 family 0.6B base	0.6B ft	1.7B base	1.7B ft	4B base	4B ft	8B base	8B ft	14B base	14B ft	Llama-3.1/3.2 family 1B base	1B ft	3B base	3B ft	8B base	8B ft	[binz2025foundation] L-70B$_p$	C$_r$	C$_p$	Cog.$_p$	Chance ln(k)
Balloon analog risk task [frey2017risk]	Decision	0.09	0.07	0.09	0.07	0.08	0.07	0.08	0.06	0.08	0.06	0.10	0.07	0.08	0.07	0.08	**0.06**	0.08	0.07	0.06	0.09	0.69
CPC18 [plonsky2018when]	Decision	0.41	0.35	0.40	0.34	0.40	0.34	0.40	0.34	0.40	**0.33**	0.43	0.36	0.41	0.35	0.39	0.34	0.41	0.35	0.34	0.66	0.69
Columbia card task [frey2017risk]	Decision	0.27	0.20	0.26	0.20	0.26	0.20	0.25	0.20	0.25	0.20	0.30	0.21	0.27	0.20	0.25	**0.20**	0.23	0.21	0.19	0.26	0.69
Decisions from description [wulff2018sampling]	Decision	1.08	0.62	1.05	0.63	1.01	0.62	0.99	0.61	1.01	0.60	1.55	0.62	1.20	0.63	0.96	0.60	0.76	**0.59**	0.53	0.61	0.69
Experiential-symbolic task [garcia2023experiential]	Decision	0.76	0.48	0.68	0.47	0.67	0.46	0.64	0.46	0.57	0.46	0.78	0.47	0.69	0.47	0.64	0.46	0.70	**0.46**	0.45	—	—
Gardening task [flesch2018comparing]	Decision	0.52	0.39	0.52	0.39	0.49	0.38	0.47	0.38	0.45	0.38	0.81	0.40	0.64	0.39	0.48	**0.38**	0.50	0.49	0.38	0.91	0.69
Go/no-go [enkavi2019large]	Decision	0.01	**0.00**	0.01	**0.00**	0.00	**0.00**	0.01	**0.00**	0.00	**0.00**	0.01	**0.00**	0.01	**0.00**	0.00	**0.00**	0.01	**0.00**	0.00	0.08	0.00
Intertemporal choice [ruggeri2022globalizability]	Decision	0.77	0.44	0.80	0.44	0.71	0.44	0.69	0.44	0.70	0.44	0.85	0.44	0.75	0.44	0.76	0.44	0.73	**0.44**	0.43	0.66	0.69
Multi-attribute DM [hilbig2014generalized]	Decision	0.49	0.09	0.34	0.08	0.20	0.09	0.19	0.08	0.19	0.08	0.63	0.08	0.45	0.08	0.23	0.08	0.15	**0.06**	0.06	0.19	0.69
Risky choice [krueger2024identifying]	Decision	0.92	0.48	0.83	0.47	0.79	0.44	0.74	0.43	0.67	**0.42**	0.99	0.49	0.85	0.46	0.76	0.43	0.65	0.43	0.43	—	—
choices13k [peterson2021using]	Decision	0.62	0.44	0.60	0.44	0.59	0.43	0.59	0.43	0.55	**0.43**	0.67	0.44	0.59	0.44	0.55	0.43	0.53	0.43	0.43	0.66	0.69
Multi-task RL [tomov2021multitask]	MDP	0.70	0.60	0.72	0.59	0.68	0.58	0.70	0.58	0.68	**0.57**	0.80	0.60	0.70	0.59	0.66	0.57	0.66	0.57	0.57	1.04	1.10
Tile-revealing task [kumar2023disentangling]	MDP	2.95	0.02	2.81	0.01	2.67	0.00	2.73	0.07	2.60	0.00	4.86	0.02	3.98	0.00	2.92	**0.00**	2.74	2.29	1.87	—	—
Two-step task [kool2017cost]	MDP	0.67	0.54	0.64	0.53	0.64	**0.53**	0.65	0.53	0.63	**0.53**	0.67	0.55	0.62	0.53	0.61	**0.53**	0.61	0.53	0.50†	0.60†	0.69
Two-step task [kool2016when]	MDP	0.69	0.51	0.64	0.50	0.64	0.49	0.65	0.49	0.63	0.48	0.67	0.52	0.62	0.50	0.61	0.49	0.61	**0.48**	0.50†	0.60†	0.69
Two-step task [zorowitz2023data]	MDP	0.62	0.52	0.59	0.51	0.60	0.51	0.61	**0.51**	0.59	0.51	0.64	0.52	0.60	0.51	0.58	0.51	0.61	0.51	0.50†	0.60†	0.69
Virtual subway network [tomov2020discovery]	MDP	1.68	1.16	1.55	1.15	1.57	1.14	1.55	1.13	1.57	**1.12**	1.68	1.19	1.60	1.16	1.53	1.13	1.53	1.16	1.13	—	1.61
Zoopermarket [ludwig2023human]	MDP	0.61	0.54	0.61	0.52	0.60	0.51	0.60	0.50	0.59	**0.48**	0.64	0.54	0.62	0.52	0.60	0.50	0.60	0.50	0.48	0.60	0.69
Cond. assoc. learning [collins2014working]	Memory	0.76	0.54	0.71	0.53	0.71	0.52	0.69	0.52	0.61	**0.51**	0.82	0.55	0.73	0.53	0.70	0.52	0.64	0.52	0.54	0.86	1.10
Digit span [enkavi2019large]	Memory	0.94	0.60	0.89	0.59	0.72	0.58	0.70	0.57	0.69	**0.57**	1.01	0.60	0.84	0.59	0.75	0.58	0.66	0.57	0.55	0.94	2.30
Episodic long-term memory [popov2023intent]	Memory	1.72	1.27	1.45	1.05	1.33	0.96	1.30	0.94	1.18	0.90	1.97	1.28	1.65	1.02	1.43	0.92	1.13	**0.89**	0.87	—	0.69
N-back [enkavi2019large]	Memory	0.58	0.42	0.58	0.42	0.55	0.41	0.55	0.41	0.53	0.40	0.59	0.42	0.57	0.42	0.57	0.41	0.52	**0.40**	0.40	0.58	0.69
Recall and recognition [cox2018information]	Memory	1.73	1.24	1.57	1.18	1.44	1.13	1.41	1.12	1.36	1.09	1.87	1.22	1.56	1.14	1.37	1.10	1.38	**1.07**	1.06	—	—
Recent probes [enkavi2019large]	Memory	0.70	0.26	0.61	0.26	0.39	0.26	0.41	0.26	0.35	0.26	0.74	0.27	0.56	0.26	0.34	**0.26**	0.34	0.26	0.26	0.39	0.69
Grammar judgement [jansen2021rational]	Misc.	2.12	1.47	2.02	1.45	2.00	1.45	1.89	1.44	1.91	1.43	2.33	1.48	2.16	1.45	2.12	1.44	1.99	1.44	1.44	**1.41**	—
Probabilistic reasoning [zhu2020bayesian]	Misc.	2.75	2.45	2.69	2.40	2.61	2.36	2.57	2.35	2.58	**2.34**	2.89	2.56	2.80	2.47	2.71	2.43	2.64	2.40	2.37	—	—
Serial reaction time task [wu2023chunking]	Misc.	0.21	0.17	0.19	0.17	0.19	0.17	0.19	0.17	0.19	**0.17**	0.31	0.17	0.23	0.17	0.20	0.17	0.19	0.18	0.17	0.20	1.39
THINGS odd-one-out [hebart2023things]	Misc.	1.21	0.82	1.21	0.81	1.17	0.80	1.15	0.79	1.13	**0.79**	1.31	0.82	1.18	0.82	1.16	0.79	1.14	0.80	0.81	0.83	1.10
Changing bandit [xiong2023neural]	Bandit	0.42	0.30	0.38	0.30	0.36	0.29	0.35	0.29	0.34	**0.29**	1.34	0.32	0.90	0.31	0.39	0.29	0.38	0.45	0.30	0.44	0.69
Decisions from experience [wulff2018sampling]	Bandit	0.74	0.49	0.66	0.49	0.69	0.49	0.78	0.49	0.61	0.48	0.81	0.50	0.71	0.49	0.65	0.48	**0.43**	0.47	0.37	0.54	—
Drifting four-armed bandit [bahrami2020four]	Bandit	0.98	0.73	0.95	0.72	0.92	0.71	0.91	0.71	0.91	**0.71**	0.95	0.73	0.94	0.72	0.90	0.71	0.88	0.71	0.70	0.90	1.39
Horizon task [feng2021dynamics]	Bandit	0.47	0.23	0.43	0.23	0.39	0.22	0.36	0.22	0.34	0.22	0.81	0.26	0.67	0.24	0.44	**0.22**	0.52	0.40	0.40†	0.36†	0.69
Horizon task [sadeghiyeh2020temporal]	Bandit	0.67	0.59	0.72	0.58	0.66	0.58	0.65	0.58	0.65	0.58	0.67	0.59	0.68	0.58	0.66	0.58	0.52	0.58	0.40†	**0.36**†	0.69
Horizon task [somerville2017charting]	Bandit	0.54	0.34	0.55	0.34	0.53	0.34	0.47	0.34	0.49	**0.33**	0.59	0.35	0.57	0.34	0.51	0.34	0.52	0.35	0.40†	0.36†	0.69
Horizon task [waltz2020differential]	Bandit	0.33	0.15	0.33	0.15	0.31	0.15	0.27	0.15	0.24	0.15	0.37	0.16	0.37	0.16	0.30	**0.14**	0.52	0.15	0.40†	0.36†	0.69
Horizon task [wilson2014humans]	Bandit	0.57	0.46	0.58	0.46	0.56	0.45	0.53	0.45	0.54	0.45	0.64	0.47	0.62	0.46	0.55	0.45	0.52	0.48	0.40†	**0.36**†	0.69
Iowa gambling task [steingroever2015data]	Bandit	1.07	0.93	1.06	0.93	1.07	0.91	1.07	0.91	1.07	0.91	1.10	0.94	1.06	0.92	1.04	0.91	0.99	**0.91**	0.89	1.16	1.39
Prob. instrumental learning [lefebvre2017behavioural]	Bandit	0.62	0.50	0.60	0.50	0.57	**0.49**	0.59	0.49	0.56	0.49	0.62	0.51	0.58	0.50	0.56	0.49	0.54	0.50	0.49	0.50	0.69
Spatially correlated MAB [wu2018generalization]	Bandit	2.61	2.01	2.55	1.96	2.50	1.93	2.44	1.87	2.46	1.86	2.76	2.16	2.64	2.00	2.58	1.93	2.45	**1.82**	1.83	2.76	3.40
Structured bandit [schulz2020finding]	Bandit	0.92	0.67	0.92	0.66	0.88	0.65	0.87	0.65	0.83	**0.64**	0.94	0.67	0.89	0.66	0.85	0.65	0.81	0.64	0.64	1.05	2.08
Two-armed bandit [gershman2018deconstructing]	Bandit	0.43	0.30	0.41	0.30	0.40	0.29	0.39	0.29	0.39	**0.29**	0.43	0.30	0.43	0.29	0.40	0.29	0.38	0.30	0.30	0.42	0.69
Aversive learning [wise2019computational]	Sup. learn.	5.20	4.71	5.06	4.55	4.95	4.27	4.89	4.24	4.75	3.81	5.66	4.58	5.52	4.80	5.25	**3.66**	5.11	4.13	4.07	—	—
Medin categorization [levering2020revisiting]	Sup. learn.	0.60	0.51	0.60	0.51	0.60	0.50	0.59	0.50	0.55	**0.50**	0.71	0.52	0.60	0.50	0.57	0.50	0.58	0.50	0.50	0.53	—
Multiple-cue judgment [collsioo2023numerical]	Sup. learn.	1.46	1.17	1.38	1.16	1.28	1.14	1.30	1.14	1.26	1.13	1.69	1.17	1.50	1.15	1.32	**1.12**	1.28	1.14	1.12	1.92	2.20
Shepard categorization [badham2017deficits]	Sup. learn.	0.63	0.55	0.61	0.55	0.59	0.54	0.58	0.53	0.59	**0.53**	0.67	0.56	0.62	0.54	0.60	0.53	0.58	0.54	0.54	0.61	0.69
Weather prediction task [speekenbrink2008learning]	Sup. learn.	0.62	0.55	0.59	0.56	0.61	0.54	0.58	0.55	0.58	0.55	0.62	0.55	0.59	**0.54**	0.58	0.56	0.57	0.56	0.55	0.63	0.69
Mean (all 48)		0.99	0.69	0.94	0.68	0.90	0.66	0.89	0.66	0.87	**0.64**	1.14	0.70	1.02	0.68	0.92	0.64	0.88	0.71	0.69	0.69	0.98

<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 1.00–1.20 1.20–1.50 1.50–2.00 >2.00

Bold+underline = best; underline = second-best (C$_p$ excluded; see footnote). NLL under bf16 (half-precision) inference. Reference columns report 4-bit values as originally published. Lower is better. "—" = no cognitive baseline reported. Size labels in billions of parameters. Subscript $_r$ denotes our reproducing evaluation of the original Centaur model under identical library and CUDA versions as our models; $_p$ denotes values published by [binz2025foundation]. C$_p$ is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. †Horizon task (5 experiments) and Two-step task (3 experiments) are reported as single merged values by [binz2025foundation]. Rounded to 2 d.p.

Table 2: Base vs. finetuned NLL on Psych-101 under bf16 inference. Within each model size, the left column (base) shows the pretrained model and the right column (ft) shows the cognitively finetuned variant. Cell colour encodes performance; the consistent colour shift from base to ft demonstrates the effect of cognitive finetuning across all scales.

	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		**Base**			[binz2025foundation]		
Task type	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	70B$_p$	70B$_r$	Cog.$_p$
Decision (11)	0.32	0.32	0.32	0.31	0.31	0.32	0.32	0.31	0.40	0.37	0.35	0.32	0.36	0.32	0.46	0.44	0.46	**0.30**	0.32	0.46
MDP (7)	0.56	0.55	0.54	0.54	**0.53**	0.56	0.54	0.53	1.03	0.91	0.59	0.54	0.63	0.56	1.07	1.04	1.07	0.79	0.86	0.69
Bandit (13)	0.59	0.59	0.58	0.57	**0.57**	0.61	0.59	0.58	0.71	0.66	0.63	0.58	0.65	0.59	0.74	0.73	0.76	0.58	0.60	0.74
Memory (6)	0.72	0.67	0.65	0.64	0.62	0.72	0.66	0.63	0.97	0.85	0.76	0.66	0.77	0.67	0.84	0.78	0.86	**0.61**	0.62	0.69
Misc. (4)	1.23	1.21	1.19	1.19	1.18	1.26	1.23	1.21	1.39	1.30	1.25	1.22	1.28	1.23	1.45	1.45	1.55	1.20	1.20	**0.81**
Sup. learn. (5)	1.50	1.47	1.40	1.39	1.30	1.48	1.51	1.27	1.74	1.66	1.59	1.33	1.75	1.48	1.59	1.55	1.66	1.36	1.37	**0.92**
Mean (46)	0.69	0.68	0.66	0.66	**0.64**	0.70	0.68	0.64	0.89	0.82	0.73	0.66	0.77	0.68	0.89	0.87	0.92	0.69	0.71	0.69

<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 >1.00

Table 3: Mean NLL by task type for finetuned models (bf16). Finetuned families: `Qwentaur`, `Llama-Centaur`, `Smoltaur`, `Olmotaur`. Base columns: `Q-8B`/`Q-14B` (Qwen3), `L-8B` (Llama-3.1). Subscript $_r$ denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript $_p$ denotes values published by [binz2025foundation]. Cell colour encodes performance; **bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix ??.

	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]		
Task type	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	70B$_p$	70B$_r$	Cog.$_p$
Decision (9)	0.29	0.29	0.29	0.28	0.28	0.29	0.29	0.28	0.36	0.33	0.32	0.28	0.33	0.29	0.41	0.40	0.41	**0.27**	0.29	0.46
MDP (5)	0.54	0.53	0.52	0.52	0.51	0.54	0.53	0.52	0.62	0.60	0.56	0.53	0.61	0.53	0.64	0.62	0.61	**0.51**	0.52	0.69
Bandit (13)	0.59	0.59	0.58	0.57	**0.57**	0.61	0.59	0.58	0.71	0.66	0.63	0.58	0.65	0.59	0.74	0.73	0.76	0.58	0.60	0.74
Memory (4)	0.45	0.45	0.44	0.44	0.44	0.46	0.45	0.44	0.57	0.51	0.48	0.45	0.49	0.45	0.59	0.54	0.59	**0.44**	0.44	0.69
Misc. (3)	0.82	0.81	0.80	0.80	**0.80**	0.83	0.81	0.80	0.96	0.87	0.84	0.81	0.84	0.81	1.08	1.08	1.16	0.80	0.81	0.81
Sup. learn. (4)	0.69	0.70	0.68	0.68	**0.68**	0.70	0.68	0.68	0.84	0.79	0.76	0.68	0.83	0.68	0.77	0.75	0.77	0.68	0.68	0.92
Mean (38)	0.53	0.52	0.52	0.51	**0.51**	0.54	0.52	0.51	0.64	0.59	0.56	0.52	0.59	0.52	0.66	0.65	0.67	0.51	0.52	0.69

<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 >1.00

Table 4: Mean NLL by task type for finetuned models (bf16), restricted to the 38 experiments (of 46) for which a cognitive model baseline is available. 8 experiments without a reported cognitive baseline are excluded to enable direct comparison across all columns. Finetuned families: `Qwentaur`, `Llama-Centaur`, `Smoltaur`, `Olmotaur`. Base columns: `Q-8B`/`Q-14B` (Qwen3), `L-8B` (Llama-3.1). Subscript $_r$ denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript $_p$ denotes values published by [binz2025foundation]. Cell colour encodes performance; **bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix **??**.

	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]			Chance
Task type	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	$70B_p$	$70B_r$	$Cog._p$	$\ln(k)$
Decision (9)	0.29	0.29	0.29	0.28	$\underline{0.28}$	0.29	0.29	0.28	0.36	0.33	0.32	0.28	0.33	0.29	0.41	0.40	0.41	$\underline{\mathbf{0.27}}$	0.29	0.46	0.62
MDP (5)	0.54	0.53	0.52	0.52	$\underline{0.51}$	0.54	0.53	0.52	0.62	0.60	0.56	0.53	0.61	0.53	0.64	0.62	0.61	$\underline{\mathbf{0.51}}$	0.52	0.69	0.77
Bandit (12)	0.60	0.60	0.59	$\underline{0.58}$	$\underline{\mathbf{0.58}}$	0.62	0.60	0.58	0.73	0.68	0.64	0.59	0.66	0.60	0.74	0.74	0.76	0.60	0.61	0.75	1.15
Memory (4)	0.45	0.45	0.44	0.44	$\underline{0.44}$	0.46	0.45	0.44	0.57	0.51	0.48	0.45	0.49	0.45	0.59	0.54	0.59	$\underline{\mathbf{0.44}}$	0.44	0.69	1.20
Misc. (2)	0.50	0.49	0.48	0.48	$\underline{\mathbf{0.48}}$	0.50	0.49	$\underline{0.48}$	0.62	0.53	0.51	0.48	0.51	0.49	0.67	0.66	0.68	0.49	0.49	0.51	1.24
Sup. learn. (3)	0.75	0.76	0.74	0.74	$\underline{\mathbf{0.74}}$	0.76	0.74	$\underline{0.74}$	0.93	0.87	0.84	0.74	0.93	0.74	0.82	0.81	0.83	$\underline{0.74}$	0.74	1.05	1.19
Mean (35)	0.50	0.50	0.49	$\underline{0.49}$	$\underline{\mathbf{0.48}}$	0.51	0.50	0.49	0.61	0.57	0.54	0.49	0.56	0.50	0.63	0.61	0.63	0.49	0.50	0.67	0.97

$<$0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 $>$1.00

Table 5: Mean NLL by task type for finetuned models (bf16), restricted to the 35 experiments (of 46) that have both a reported cognitive model baseline and a well-defined discrete response space admitting a chance-level baseline $\ln(k)$, where k is the number of response options. 11 experiments lacking either a cognitive baseline or a computable $\ln(k)$ (continuous, mixed, or variable action spaces) are excluded. Finetuned families: `Qwentaur`, `Llama-Centaur`, `Smoltaur`, `Olmotaur`. Base columns: `Q-8B`/`Q-14B` (Qwen3), `L-8B` (Llama-3.1). Subscript $_r$ denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript $_p$ denotes values published by [binz2025foundation]. Cell colour encodes performance; $\underline{\textbf{bold+underline}}$ marks the best model, $\underline{\text{underline}}$ the second-best. Full per-experiment results in Appendix **??**.

Task type	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]		
	`0.6B`	`1.7B`	`4B`	`8B`	`14B`	`1B`	`3B`	`8B`	`0.1B`	`0.4B`	`1.7B`	`3B`	`1B`	`7B`	`Q-8B`	`Q-14B`	`L-8B`	70B_p	70B_r	Cog._p
Decision (8)	0.53	0.53	0.54	0.54	<u>0.54</u>	0.53	0.53	0.54	0.41	0.46	0.48	0.54	0.47	0.53	0.34	0.35	0.33	**<u>0.56</u>**	0.52	0.27
MDP (5)	0.28	0.30	0.31	0.31	<u>0.32</u>	0.28	0.30	0.31	0.18	0.20	0.26	0.30	0.20	0.30	0.15	0.17	0.19	**<u>0.32</u>**	0.31	0.11
Bandit (12)	0.48	0.48	0.49	<u>0.49</u>	**<u>0.50</u>**	0.46	0.48	0.49	0.36	0.41	0.44	0.49	0.42	0.48	0.35	0.36	0.33	0.46	0.45	0.39
Memory (4)	0.57	0.57	0.58	0.58	0.58	0.56	0.57	0.58	0.46	0.51	0.55	0.57	0.53	0.57	0.42	0.47	0.43	<u>0.58</u>	**<u>0.58</u>**	0.36
Misc. (2)	0.57	0.57	0.58	0.58	**<u>0.58</u>**	0.56	0.57	<u>0.58</u>	0.45	0.53	0.55	0.58	0.55	0.57	0.41	0.42	0.40	0.57	0.57	0.55
Sup. learn. (3)	0.30	0.29	0.31	0.31	<u>0.31</u>	0.29	0.31	0.31	0.18	0.23	0.25	0.30	0.20	**<u>0.31</u>**	0.24	0.24	0.23	0.30	0.30	0.11
Mean (34)	0.46	0.46	0.47	0.48	**<u>0.48</u>**	0.45	0.47	<u>0.48</u>	0.35	0.39	0.43	0.47	0.40	0.47	0.32	0.34	0.32	0.47	0.46	0.30

<0.05 0.05–0.12 0.12–0.18 0.18–0.25 0.25–0.32 0.32–0.40 0.40–0.48 0.48–0.55 ≥ 0.55

Table 6: Fraction of available information captured above chance, $(\ln k - \text{NLL}) / \ln k$, by task type for finetuned models (bf16). A value of 0 indicates chance-level performance; 1 indicates perfect prediction. Restricted to 34 experiments (of 46) with both a cognitive model baseline and a well-defined discrete response space ($\ln k > 0$); 12 experiments are excluded. Finetuned families: `Qwentaur`, `Llama-Centaur`, `Smoltaur`, `Olmotaur`. Base columns: `Q-8B`/`Q-14B` (Qwen3), `L-8B` (Llama-3.1). Subscript $_r$ denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript $_p$ denotes values published by [binz2025foundation]. **<u>Bold+underline</u>** marks the best model, <u>underline</u> the second-best. Full per-experiment results in Appendix **??**.

Experiment	Type	Cognitive SFT					Other behavioural FMs			Non-cognitive finetuned					Chance
		Q-8B	Q-14B	LC-8B	C-70B$_r$	Cog.$_p$	Be.FM-8B	Socr.-8B	Socr.-14B	Herm.-3B	Herm.-8B	Herm.-14B	Nemo.-4B	Nemo.-8B	ln(k)
Balloon analog risk task [frey2017risk]	Decision	0.06	0.06	**0.06**	0.07	0.09	—	13.52	5.87	0.09	0.09	0.08	0.17	0.18	0.69
CPC18 [plonsky2018when]	Decision	0.34	**0.33**	0.34	0.35	0.66	—	14.34	8.54	0.44	0.42	0.40	0.95	0.64	0.69
Columbia card task [frey2017risk]	Decision	0.20	0.20	**0.20**	0.21	0.26	—	14.07	6.72	0.27	0.26	0.26	0.48	0.49	0.69
Decisions from description [wulff2018sampling]	Decision	0.61	0.60	0.60	**0.59**	0.61	—	13.59	10.13	1.34	1.17	1.02	2.99	4.44	0.69
Experiential-symbolic task [garcia2023experiential]	Decision	0.46	0.46	0.46	**0.46**	—	—	12.95	7.86	0.79	0.67	0.58	1.37	0.95	—
Gardening task [flesch2018comparing]	Decision	0.38	0.38	**0.38**	0.49	0.91	—	16.26	10.28	0.53	0.46	0.45	0.61	0.64	0.69
Go/no-go [enkavi2019large]	Decision	**0.00**	**0.00**	**0.00**	**0.00**	0.08	—	12.55	4.24	0.01	0.00	0.00	0.04	0.01	0.00
Intertemporal choice [ruggeri2022globalizability]	Decision	0.44	0.44	0.44	**0.44**	0.66	—	13.82	12.23	0.72	1.04	0.74	2.75	2.72	0.69
Multi-attribute DM [hilbig2014generalized]	Decision	0.08	0.08	0.08	**0.06**	0.19	—	13.86	10.45	0.45	0.23	0.20	0.61	0.54	0.69
Risky choice [krueger2024identifying]	Decision	0.43	**0.42**	0.43	0.43	—	—	16.10	7.17	0.94	0.82	0.67	1.79	1.35	—
choices13k [peterson2021using]	Decision	0.43	**0.43**	0.43	0.43	0.66	—	13.44	9.40	0.65	0.60	0.53	1.65	1.13	0.69
Multi-task RL [tomov2021multitask]	MDP	0.58	**0.57**	0.57	0.57	1.04	—	14.70	7.95	0.81	0.70	0.66	1.20	1.06	1.10
Tile-revealing task [kumar2023disentangling]	MDP	0.07	0.00	**0.00**	2.29	—	—	65.75	35.15	2.93	2.85	2.62	3.48	3.37	—
Two-step task [kool2017cost]	MDP	0.53	**0.53**	**0.53**	0.53	0.60	—	14.06	7.90	0.67	0.67	0.63	1.17	0.81	0.69
Two-step task [kool2016when]	MDP	0.49	0.48	0.49	**0.48**	0.60	—	14.63	8.82	0.68	0.67	0.63	1.15	0.82	0.69
Two-step task [zorowitz2023data]	MDP	**0.51**	0.51	0.51	0.51	0.60	—	13.18	7.68	0.63	0.62	0.58	1.03	0.82	0.69
Virtual subway network [tomov2020discovery]	MDP	1.13	**1.12**	1.13	1.16	—	—	18.21	11.39	1.69	1.64	1.60	2.41	2.10	1.61
Zoopermarket [ludwig2023human]	MDP	0.50	**0.48**	0.50	0.50	0.60	—	13.75	6.62	0.65	0.63	0.59	1.49	1.09	0.69
Cond. assoc. learning [collins2014working]	Memory	0.52	**0.51**	0.52	0.52	0.86	—	16.16	10.04	0.80	0.68	0.59	0.92	0.95	1.10
Digit span [enkavi2019large]	Memory	0.57	**0.57**	0.58	0.57	0.94	—	4.65	2.96	0.87	0.83	0.70	2.16	1.02	2.30
Episodic long-term memory [popov2023intent]	Memory	0.94	0.90	0.92	**0.89**	—	—	17.12	9.08	1.82	1.72	1.21	3.07	2.22	0.69
N-back [enkavi2019large]	Memory	0.41	0.40	0.41	**0.40**	0.58	—	14.42	9.24	0.59	0.57	0.53	0.79	0.75	0.69
Recall and recognition [cox2018information]	Memory	1.12	1.09	1.10	**1.07**	—	—	24.31	9.33	1.65	1.61	1.38	4.16	2.60	—
Recent probes [enkavi2019large]	Memory	0.26	0.26	**0.26**	0.26	0.39	—	14.16	9.92	0.61	0.39	0.34	1.00	1.06	0.69
Grammar judgement [jansen2021rational]	Misc.	1.44	1.43	1.44	1.44	**1.41**	—	12.88	10.37	2.27	2.53	1.94	3.90	3.12	—
Probabilistic reasoning [zhu2020bayesian]	Misc.	2.35	**2.34**	2.43	2.40	—	—	3.40	2.92	2.98	2.97	2.59	4.05	3.23	—
Serial reaction time task [wu2023chunking]	Misc.	0.17	**0.17**	0.17	0.18	0.20	—	14.43	5.09	0.21	0.20	0.19	0.25	0.26	1.39
THINGS odd-one-out [hebart2023things]	Misc.	0.79	**0.79**	0.79	0.80	0.83	—	14.31	12.15	1.24	1.28	1.15	2.13	1.75	1.10
Changing bandit [xiong2023neural]	Bandit	0.29	**0.29**	0.29	0.45	0.44	—	14.76	7.52	0.39	0.36	0.34	0.78	0.53	0.69
Decisions from experience [wulff2018sampling]	Bandit	0.49	0.48	0.48	**0.47**	0.54	—	13.74	7.18	0.69	0.88	0.64	2.29	1.78	—
Drifting four-armed bandit [bahrami2020four]	Bandit	0.71	**0.71**	0.71	0.71	0.90	—	13.91	8.43	0.99	0.96	0.90	2.10	1.78	1.39
Horizon task [feng2021dynamics]	Bandit	0.22	0.22	**0.22**	0.40	0.36	—	15.10	7.55	0.43	0.36	0.35	0.65	0.62	0.69
Horizon task [sadeghiyeh2020temporal]	Bandit	0.58	0.58	0.58	0.58	**0.36**	—	14.94	8.62	0.73	0.69	0.66	1.10	0.93	0.69
Horizon task [somerville2017charting]	Bandit	0.34	**0.33**	0.34	0.35	0.36	—	15.18	8.49	0.57	0.47	0.49	0.86	0.74	0.69
Horizon task [waltz2020differential]	Bandit	0.15	0.15	**0.14**	0.15	0.36	—	15.42	7.82	0.35	0.26	0.26	0.54	0.54	0.69
Horizon task [wilson2014humans]	Bandit	0.45	0.45	0.45	0.48	**0.36**	—	15.39	8.06	0.62	0.56	0.54	0.89	0.79	0.69
Iowa gambling task [steingroever2015data]	Bandit	0.91	0.91	0.91	**0.91**	1.16	—	13.23	8.45	1.13	1.09	1.06	2.27	1.60	1.39
Prob. instrumental learning [lefebvre2017behavioural]	Bandit	**0.49**	0.49	0.49	0.50	0.50	—	15.92	8.30	0.68	0.54	0.53	1.16	0.86	0.69
Spatially correlated MAB [wu2018generalization]	Bandit	1.87	1.86	1.93	**1.82**	2.76	—	5.43	3.14	2.64	2.78	2.48	3.93	3.19	3.40
Structured bandit [schulz2020finding]	Bandit	0.65	**0.64**	0.65	0.64	1.05	—	2.12	1.57	0.91	0.90	0.80	1.41	1.08	2.08
Two-armed bandit [gershman2018deconstructing]	Bandit	0.29	**0.29**	0.29	0.30	0.42	—	13.99	7.30	0.47	0.44	0.41	0.98	0.73	0.69
Aversive learning [wise2019computational]	Sup. learn.	4.24	3.81	**3.66**	4.13	—	—	18.42	7.19	5.52	5.33	4.78	7.03	6.14	—
Medin categorization [levering2020revisiting]	Sup. learn.	0.50	**0.50**	0.50	0.50	0.53	—	13.53	11.00	0.63	0.62	0.56	1.15	0.90	—
Multiple-cue judgment [collsioo2023numerical]	Sup. learn.	1.14	1.13	**1.12**	1.14	1.92	—	18.06	6.92	1.51	1.37	1.27	2.56	1.85	2.20
Shepard categorization [badham2017deficits]	Sup. learn.	0.53	**0.53**	0.53	0.54	0.61	—	12.66	10.13	0.64	0.63	0.59	0.92	0.76	0.69
Weather prediction task [speekenbrink2008learning]	Sup. learn.	**0.55**	0.55	0.56	0.56	0.63	—	17.01	10.65	0.63	0.61	0.56	0.89	0.77	0.69
Mean (all 46)		0.66	**0.64**	0.64	0.71	0.69	—	15.07	8.65	1.02	0.98	0.87	1.72	1.43	0.98

<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 1.00–1.20 1.20–1.50 1.50–2.00 >2.00

Bold+underline = best; underline = second-best. Lower is better. "—" = not available. Q = Qwentaur; LC = Llama-Centaur; C-70B$_r$ = Centaur-70B (reproduced); Cog.$_p$ = best cognitive model [binz2025foundation]; Herm. = Hermes [teknium2024hermes]; Nemo. = Nemotron [nvidia2025nemotron]; Socr. = Socrates [kolluri2025finetuning]. Cognitive SFT models are finetuned on behavioural data (Psych-101 or domain-specific); non-cognitive models are finetuned for general instruction-following or reasoning tasks.

Table 7: Comparison of cognitively finetuned models with other behavioural foundation models and non-cognitive controls on Psych-101. Non-cognitive finetuned models (Hermes, Nemotron) are instruction-tuned or reasoning-optimised variants of the same base model families. Their substantially higher NLL demonstrates that finetuning on non-cognitive data does not align models with human behavioural patterns, ruling out the hypothesis that any finetuning improves cognitive prediction.

Experiment	Type	Qwen 2.5/3 family: Base 8B	Base 14B	Qwentaur 8B	Qwentaur 14B	Other cog. Socr.-14B	Non-cog. Herm.-14B	Llama 3/3.1/3.2 family: Base 3B	Base 8B	Llama-Centaur 3B	Llama-Centaur 8B	Other cog. Be.FM-8B	Other cog. Socr.-8B	Non-cog. Herm.-3B	Nemo.-4B	Herm.-8B	Nemo.-8B	[binz2025foundation] C-70B$_r$	Cog.$_p$	Chance ln(k)
Balloon analog risk task [frey2017risk]	Decision	0.08	0.08	0.06	0.06	5.87	0.08	0.08	0.08	0.07	**0.06**	—	13.52	0.09	0.17	0.09	0.18	0.07	0.09	0.69
CPC18 [plonsky2018when]	Decision	0.40	0.40	0.34	**0.33**	8.54	0.40	0.41	0.39	0.35	0.34	—	14.34	0.44	0.95	0.42	0.64	0.35	0.66	0.69
Columbia card task [frey2017risk]	Decision	0.25	0.25	0.20	0.20	6.72	0.26	0.27	0.25	0.20	**0.20**	—	14.07	0.27	0.48	0.26	0.49	0.21	0.26	0.69
Decisions from description [wulff2018sampling]	Decision	0.99	1.01	0.61	0.60	10.13	1.02	1.20	0.96	0.63	0.60	—	13.59	1.34	2.99	1.17	4.44	**0.59**	0.61	0.69
Experiential-symbolic task [garcia2023experiential]	Decision	0.64	0.57	0.46	0.46	7.86	0.58	0.69	0.64	0.47	0.46	—	12.95	0.79	1.37	0.67	0.95	**0.46**	—	—
Gardening task [flesch2018comparing]	Decision	0.47	0.45	0.38	0.38	10.28	0.45	0.64	0.48	0.39	**0.38**	—	16.26	0.53	0.61	0.46	0.64	0.49	0.91	0.69
Go/no-go [enkavi2019large]	Decision	0.01	0.00	**0.00**	**0.00**	4.24	0.00	0.01	0.00	**0.00**	**0.00**	—	12.55	0.01	0.04	0.00	0.01	**0.00**	0.08	0.00
Intertemporal choice [ruggeri2022globalizability]	Decision	0.69	0.70	0.44	0.44	12.23	0.74	0.75	0.76	0.44	0.44	—	13.82	0.72	2.75	1.04	2.72	**0.44**	0.66	0.69
Multi-attribute DM [hilbig2014generalized]	Decision	0.19	0.19	0.08	0.08	10.45	0.20	0.45	0.23	0.08	0.08	—	13.86	0.45	0.61	0.23	0.54	**0.06**	0.19	0.69
Risky choice [krueger2024identifying]	Decision	0.74	0.67	0.43	**0.42**	7.17	0.67	0.85	0.76	0.46	0.43	—	16.10	0.94	1.79	0.82	1.35	0.43	—	—
choices13k [peterson2021using]	Decision	0.59	0.55	0.43	**0.43**	9.40	0.53	0.59	0.55	0.44	0.43	—	13.44	0.65	1.65	0.60	1.13	0.43	0.66	0.69
Multi-task RL [tomov2021multitask]	MDP	0.70	0.68	0.58	**0.57**	7.95	0.66	0.70	0.66	0.59	0.57	—	14.70	0.81	1.20	0.70	1.06	0.57	1.04	1.10
Tile-revealing task [kumar2023disentangling]	MDP	2.73	2.60	0.07	0.00	35.15	2.62	3.98	2.92	0.00	**0.00**	—	65.75	2.93	3.48	2.85	3.37	2.29	—	—
Two-step task [kool2017cost]	MDP	0.65	0.63	0.53	**0.53**	7.90	0.63	0.62	0.61	0.53	**0.53**	—	14.06	0.67	1.17	0.67	0.81	0.53	0.60	0.69
Two-step task [kool2016when]	MDP	0.65	0.63	0.49	0.48	8.82	0.63	0.62	0.61	0.50	0.49	—	14.63	0.68	1.15	0.67	0.82	**0.48**	0.60	0.69
Two-step task [zorowitz2023data]	MDP	0.61	0.59	**0.51**	0.51	7.68	0.58	0.60	0.58	0.51	0.51	—	13.18	0.63	1.03	0.62	0.82	0.51	0.60	0.69
Virtual subway network [tomov2020discovery]	MDP	1.55	1.57	1.13	**1.12**	11.39	1.60	1.60	1.53	1.16	1.13	—	18.21	1.69	2.41	1.64	2.10	1.16	—	1.61
Zoopermarket [ludwig2023human]	MDP	0.60	0.59	0.50	**0.48**	6.62	0.59	0.62	0.60	0.52	0.50	—	13.75	0.65	1.49	0.63	1.09	0.50	0.60	0.69
Cond. assoc. learning [collins2014working]	Memory	0.69	0.61	0.52	**0.51**	10.04	0.59	0.73	0.70	0.53	0.52	—	16.16	0.80	0.92	0.68	0.95	0.52	0.86	1.10
Digit span [enkavi2019large]	Memory	0.70	0.69	0.57	**0.57**	2.96	0.70	0.84	0.75	0.59	0.58	—	4.65	0.87	2.16	0.83	1.02	0.57	0.94	2.30
Episodic long-term memory [popov2023intent]	Memory	1.30	1.18	0.94	0.90	9.08	1.21	1.65	1.43	1.02	0.92	—	17.12	1.82	3.07	1.72	2.22	**0.89**	—	0.69
N-back [enkavi2019large]	Memory	0.55	0.53	0.41	0.40	9.24	0.53	0.57	0.57	0.42	0.41	—	14.42	0.59	0.79	0.57	0.75	**0.40**	0.58	0.69
Recall and recognition [cox2018information]	Memory	1.41	1.36	1.12	1.09	9.33	1.38	1.56	1.37	1.14	1.10	—	24.31	1.65	4.16	1.61	2.60	**1.07**	—	—
Recent probes [enkavi2019large]	Memory	0.41	0.35	0.26	0.26	9.92	0.34	0.56	0.34	0.26	**0.26**	—	14.16	0.61	1.00	0.39	1.06	0.26	0.39	0.69
Grammar judgement [jansen2021rational]	Misc.	1.89	1.91	1.44	1.43	10.37	1.94	2.16	2.12	1.45	1.44	—	12.88	2.27	3.90	2.53	3.12	1.44	**1.41**	—
Probabilistic reasoning [zhu2020bayesian]	Misc.	2.57	2.58	2.35	**2.34**	2.92	2.59	2.80	2.71	2.47	2.43	—	3.40	2.98	4.05	2.97	3.23	2.40	—	—
Serial reaction time task [wu2023chunking]	Misc.	0.19	0.19	0.17	**0.17**	5.09	0.19	0.23	0.20	0.17	0.17	—	14.43	0.21	0.25	0.20	0.26	0.18	0.20	1.39
THINGS odd-one-out [hebart2023things]	Misc.	1.15	1.13	0.79	**0.79**	12.15	1.15	1.18	1.16	0.82	0.79	—	14.31	1.24	2.13	1.28	1.75	0.80	0.83	1.10
Changing bandit [xiong2023neural]	Bandit	0.35	0.34	0.29	**0.29**	7.52	0.34	0.90	0.39	0.31	0.29	—	14.76	0.39	0.78	0.36	0.53	0.45	0.44	0.69
Decisions from experience [wulff2018sampling]	Bandit	0.78	0.61	0.49	0.48	7.18	0.64	0.71	0.65	0.49	0.48	—	13.74	0.69	2.29	0.88	1.78	**0.47**	0.54	—
Drifting four-armed bandit [bahrami2020four]	Bandit	0.91	0.91	0.71	**0.71**	8.43	0.90	0.94	0.90	0.72	0.71	—	13.91	0.99	2.10	0.96	1.78	0.71	0.90	1.39
Horizon task [feng2021dynamics]	Bandit	0.36	0.34	0.22	0.22	7.55	0.35	0.67	0.44	0.24	**0.22**	—	15.10	0.43	0.65	0.36	0.62	0.40	0.36	0.69
Horizon task [sadeghiyeh2020temporal]	Bandit	0.65	0.65	0.58	0.58	8.62	0.66	0.68	0.66	0.58	0.58	—	14.94	0.73	1.10	0.69	0.93	0.58	**0.36**	0.69
Horizon task [somerville2017charting]	Bandit	0.47	0.49	0.34	**0.33**	8.49	0.49	0.57	0.51	0.34	0.34	—	15.18	0.57	0.86	0.47	0.74	0.35	0.36	0.69
Horizon task [waltz2020differential]	Bandit	0.27	0.24	0.15	0.15	7.82	0.26	0.37	0.30	0.16	**0.14**	—	15.42	0.35	0.54	0.26	0.54	0.15	0.36	0.69
Horizon task [wilson2014humans]	Bandit	0.53	0.54	0.45	0.45	8.06	0.54	0.62	0.55	0.46	0.45	—	15.39	0.62	0.89	0.56	0.79	0.48	**0.36**	0.69
Iowa gambling task [steingroever2015data]	Bandit	1.07	1.07	0.91	0.91	8.45	1.06	1.06	1.04	0.92	0.91	—	13.23	1.13	2.27	1.09	1.60	**0.91**	1.16	1.39
Prob. instrumental learning [lefebvre2017behavioural]	Bandit	0.59	0.56	**0.49**	0.49	8.30	0.53	0.58	0.56	0.50	0.49	—	15.92	0.68	1.16	0.54	0.86	0.50	0.50	0.69
Spatially correlated MAB [wu2018generalization]	Bandit	2.44	2.46	1.87	1.86	3.14	2.48	2.64	2.58	2.00	1.93	—	5.43	2.64	3.93	2.78	3.19	**1.82**	2.76	3.40
Structured bandit [schulz2020finding]	Bandit	0.87	0.83	0.65	**0.64**	1.57	0.80	0.89	0.85	0.66	0.65	—	2.12	0.91	1.41	0.90	1.08	0.64	1.05	2.08
Two-armed bandit [gershman2018deconstructing]	Bandit	0.39	0.39	0.29	**0.29**	7.30	0.41	0.43	0.40	0.29	0.29	—	13.99	0.47	0.98	0.44	0.73	0.30	0.42	0.69
Aversive learning [wise2019computational]	Sup. learn.	4.89	4.75	4.24	3.81	7.19	4.78	5.52	5.25	4.80	**3.66**	—	18.42	5.52	7.03	5.33	6.14	4.13	—	—
Medin categorization [levering2020revisiting]	Sup. learn.	0.59	0.55	0.50	**0.50**	11.00	0.56	0.60	0.57	0.50	0.50	—	13.53	0.63	1.15	0.62	0.90	0.50	0.53	—
Multiple-cue judgment [collsioo2023numerical]	Sup. learn.	1.30	1.26	1.14	1.13	6.92	1.27	1.50	1.32	1.15	**1.12**	—	18.06	1.51	2.56	1.37	1.85	1.14	1.92	2.20
Shepard categorization [badham2017deficits]	Sup. learn.	0.58	0.59	0.53	**0.53**	10.13	0.59	0.62	0.60	0.54	0.53	—	12.66	0.64	0.92	0.63	0.76	0.54	0.61	0.69
Weather prediction task [speekenbrink2008learning]	Sup. learn.	0.58	0.58	0.55	0.55	10.65	0.56	0.59	0.58	**0.54**	0.56	—	17.01	0.63	0.89	0.61	0.77	0.56	0.63	0.69
Mean (all 46)		0.89	0.87	0.66	**0.64**	8.65	0.87	1.02	0.92	0.68	0.64	—	15.07	1.02	1.72	0.98	1.43	0.71	0.69	0.98

<0.20 | 0.20–0.40 | 0.40–0.60 | 0.60–0.80 | 0.80–1.00 | 1.00–1.20 | 1.20–1.50 | 1.50–2.00 | >2.00

Bold+underline = best; underline = second-best. Lower is better. "—" = not available. Models are grouped by base-model family. "Base" = pretrained model without finetuning (bf16); "Other cog." = other cognitive/behavioural foundation models finetuned on non-Psych-101 behavioural data; "Non-cog." = models finetuned for general instruction-following or reasoning. `Herm.` = Hermes [teknium2024hermes]; `Nemo.` = Nemotron [nvidia2025nemotron]; `Socr.` = Socrates [kolluri2025finetuning]; `Be.FM` = [xie2025fm]. Base model versions differ within families: `Qwentaur` uses Qwen3; `Socr.-14B` uses Qwen2.5; `Herm.-14B` uses Qwen2.5. `Llama-Centaur` uses Llama-3.1/3.2; `Socr.-8B` uses Llama-3; `Be.FM` uses Llama-3.1.

Table 8: Comparison of cognitively finetuned models with other behavioural foundation models and non-cognitive controls on Psych-101, grouped by base-model family. Within each family, `Qwentaur`/`Llama-Centaur` (cognitive SFT on Psych-101) consistently outperform both other behavioural FMs and non-cognitive controls, demonstrating that alignment with human behavioural patterns requires cognitive finetuning on structured experimental data, not merely finetuning per se.